

**CSC 226 Exercise 4**  
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**Idea:**

Think of each set as a connected component of an undirected graph like in the **Connected Components** motivation for disjoint sets. We can store the graph on vertices  $0, 1, \dots, n-1$  using an adjacency matrix  $A_{n \times n}$  (i.e.,  $A[i][j] = 1$  if there is an edge between vertices  $i$  and  $j$ , and  $A[i][j] = 0$  otherwise).

**Union:**

To perform  $\text{Union}(x, y)$ , we simply set  $A[x][y] = 1$  and  $A[y][x] = 1$  in constant time. This is obtained from the observation that when a new edge  $uv$  is added to the graph, either  $u$  and  $v$  are in the said component, or the components containing  $u$  and  $v$  will be merged into a single component once the edge  $uv$  is added. Therefore, the Union operation can be implemented in worst-case constant time.

**Find:**

We know that  $\text{Find}(x)$  returns the designated representative of the set containing  $x$ . Here, the set containing  $x$  is the connected component of the graph that contains vertex  $x$ . To find the representative of this set, we can perform a breadth-first search (BFS) or depth-first search (DFS) starting from vertex  $x$  to find all vertices in the same connected component. We can then return the smallest vertex in this component as the representative. Since we may need to visit all vertices, the running time of  $\text{Find}(x)$  is  $O(n^2)$  in the worst case, because with an adjacency matrix we must scan  $n$  entries to find the neighbors of each of up to  $n$  visited vertices, where  $n$  is the number of elements in the set. This running time depends only on  $n$  and not on any other parameter, such as the number of Union operations performed so far.

**Thus,**

$\text{Union}(x, y)$  takes  $O(1)$  worst-case time.  $\text{Find}(x)$  takes  $O(n^2)$  worst-case time.